

DATA MODEL & ERD

The most common types of data models include the relational model, and entity relationship data model.

Entity-Relationship Diagram (ERD) represents entities and their relationships: An Entity Relationship Data model offers an alternative to the relational data model. The entity relationship model suggests conceptualizing a database as a collection of entities and objects existing independently within the database.

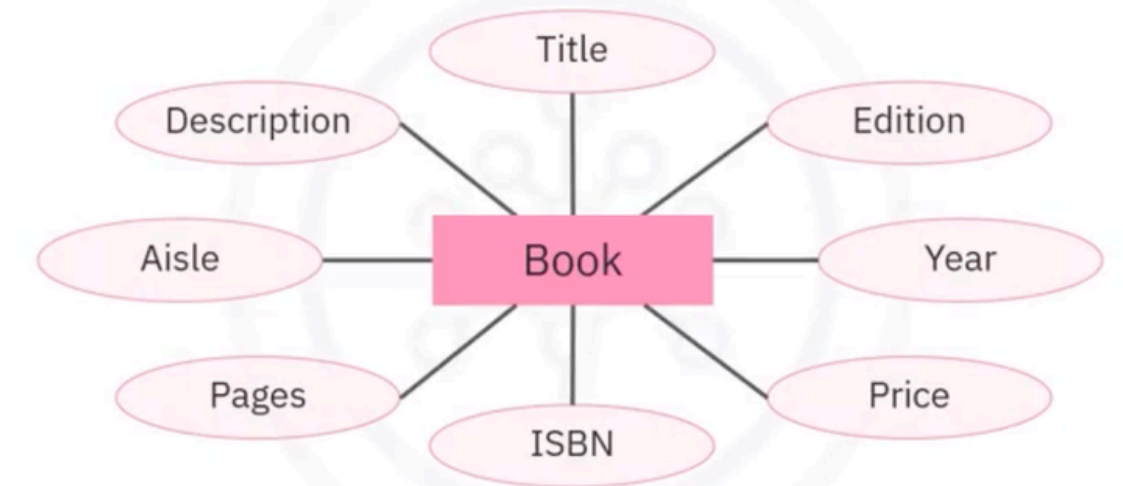
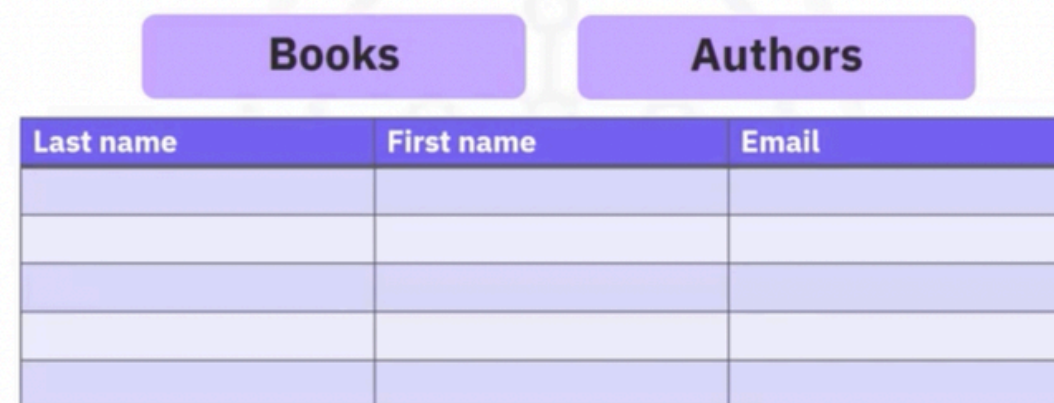
Converting an ERD into tables is easy. Entity becomes a table in the database, with attributes as columns.

Entities are in rectangles. Attributes are in ovals. Each attribute connects to exactly one entity.

- Building blocks: entities and attributes



- Building blocks: entities and attributes



DATA MODEL & ERD

The relationship is about entities.

One-to-one relationship



- Book written by a single author
- Each entity is engaged in one relationship

One-to-many relationship



- Book written by multiple authors
- Crow's foot notation is the less-than symbol (<)
- Many-to-one relationship
 - For many authors, writing a single book

Many-to-many relationship

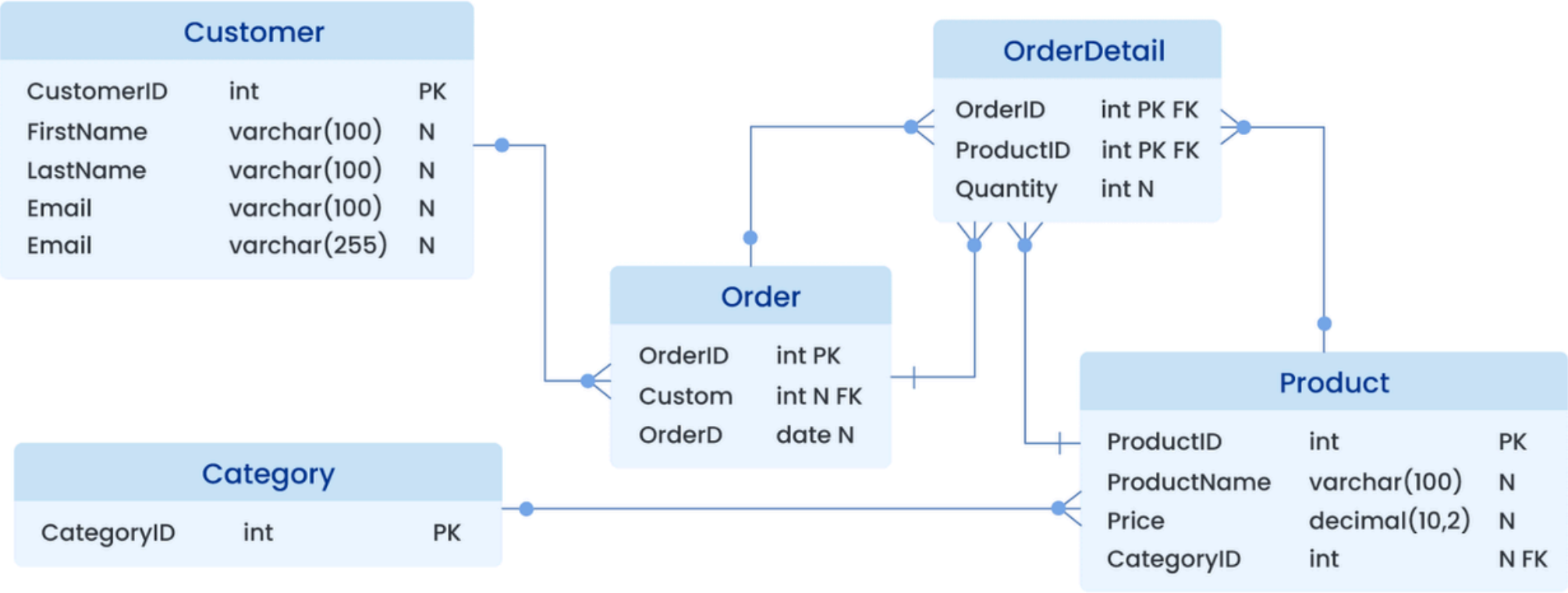


- Many authors writing many books
- Use:
 - Greater-than (>)
 - Less-than (<)
- Multiple relationships involve each entity

RELATION

Relation includes schema and instance

Schema



Instance

Relational instance of SALE:

Salesperson_id	Serial_no	Date	Sale_price
10001	1we4ds87	12/03/2020	\$ 10,000.00
10005	d63jw3ty	12/03/2020	\$ 5,000.00
10009	sy63bjd1	13/03/2020	\$ 25,000.00
10001	k2k4edr8	13/03/2020	\$ 49,000.00
10051	w3r334ac	13/03/2020	\$ 8,000.00

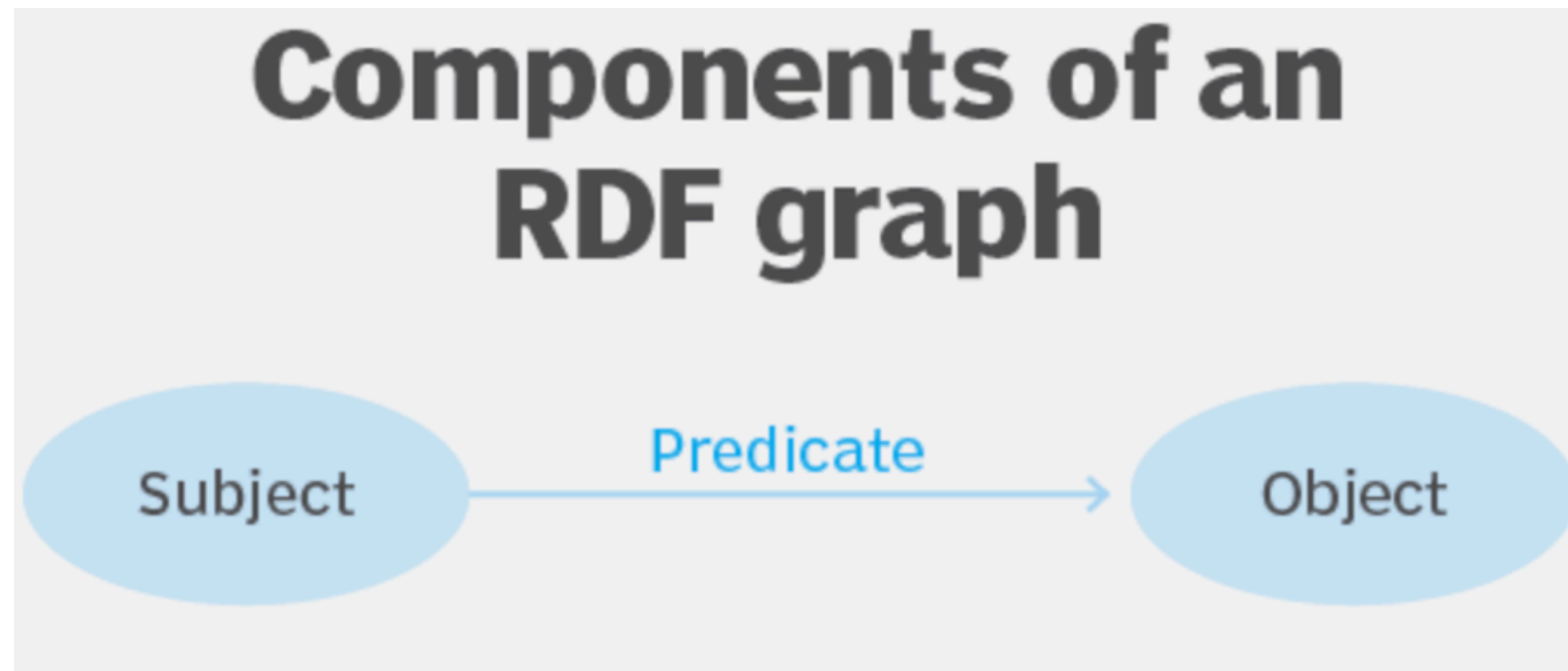
RDF

Three elements together form a triplet.

The resource that the triple is describing is the subject.

The predicate indicates the relationship between the subject and the object.

A resource that is related to the topic under discussion is called an object



ONTOLOGY

Ontology:

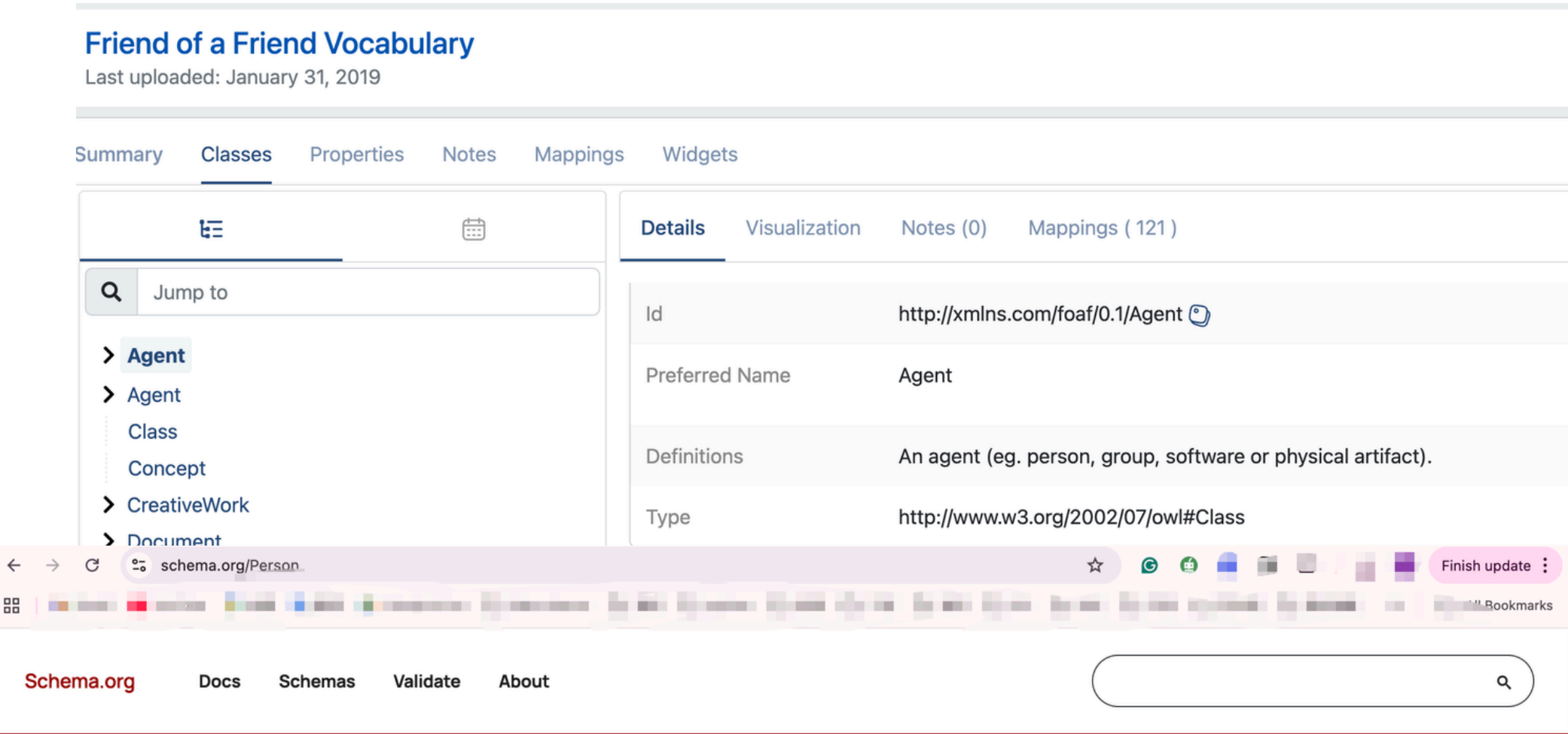
- knowledge of a domain.
- Hosted on server.
- Class in ontology can be accessed via URL.

Class: entity

Object property: defines the relationship between classes

Data property: links classes to literals such as string, integers and so on

Individual: cell value



Person

A Schema.org Type

Thing > Person

A person (alive, dead, undead, or fictional).

[more...]

Property	Expected Type	Description
Properties from Person		
additionalName	Text	An additional name for a Person, can be used for a middle name.
address	PostalAddress or Text	Physical address of the item.
affiliation	Organization	An organization that this person is affiliated with. For example, a school/university, a club, or a team.
agentInteractionStatistic	InteractionCounter	The number of completed interactions for this entity, in a particular role (the 'agent'), in a particular action (indicated in the statistic), and in a particular context (i.e. interactionService).

CDM

Provide the structure of your RDF

CDM:

- Is the Data Model targeted to your data.
- Can be regarded as a ‘special ontology’ for the sub- community or project.
- IRI/URI of a class cannot be accessed on the web; it is unique in your own ontology, unless you host your ontology on a server.

Mapping:

Class in CDM mapped to class in ontology to ensure system interoperability (optional)

owl:equivalentClass,
skos:exactMatch,
owl:sameAs...

TABLE TO RDF

Map your table to your CDM before the automatic transformation

Relational table: patient

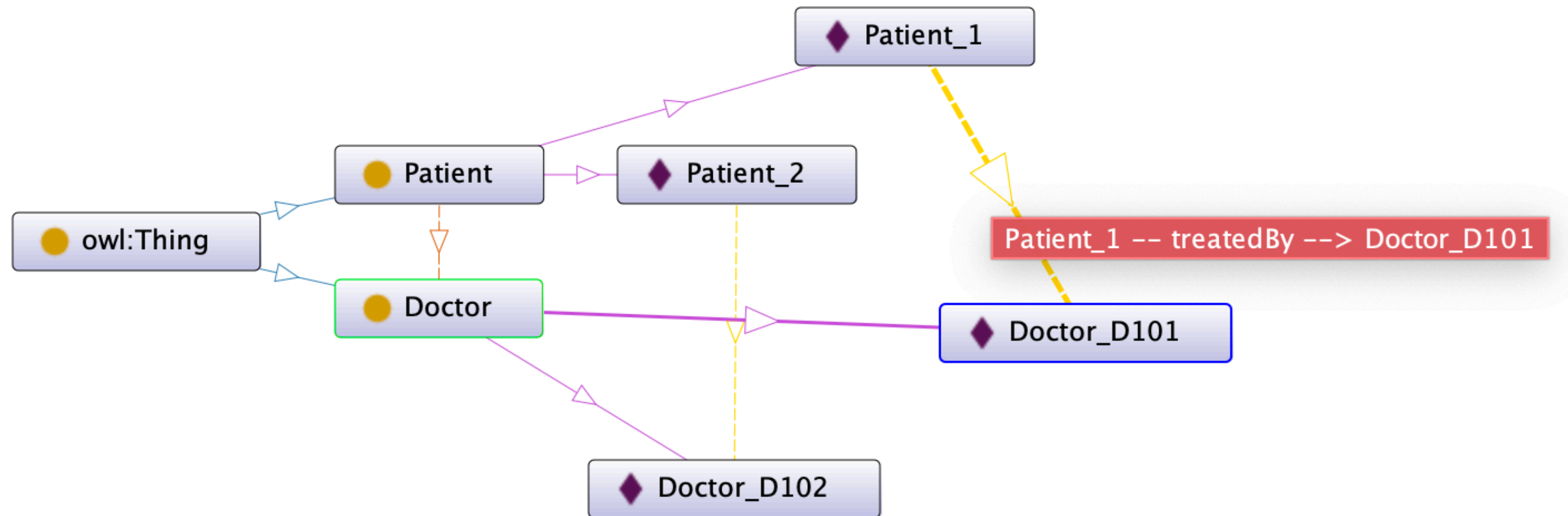
PatientID	Name	Age	DoctorID
1	Alice	30	D101
2	Bob	45	D102

Relational table: doctor

DoctorID	Name	Department
D101	Dr. Smith	Oncology
D102	Dr. Brown	Pediatrics

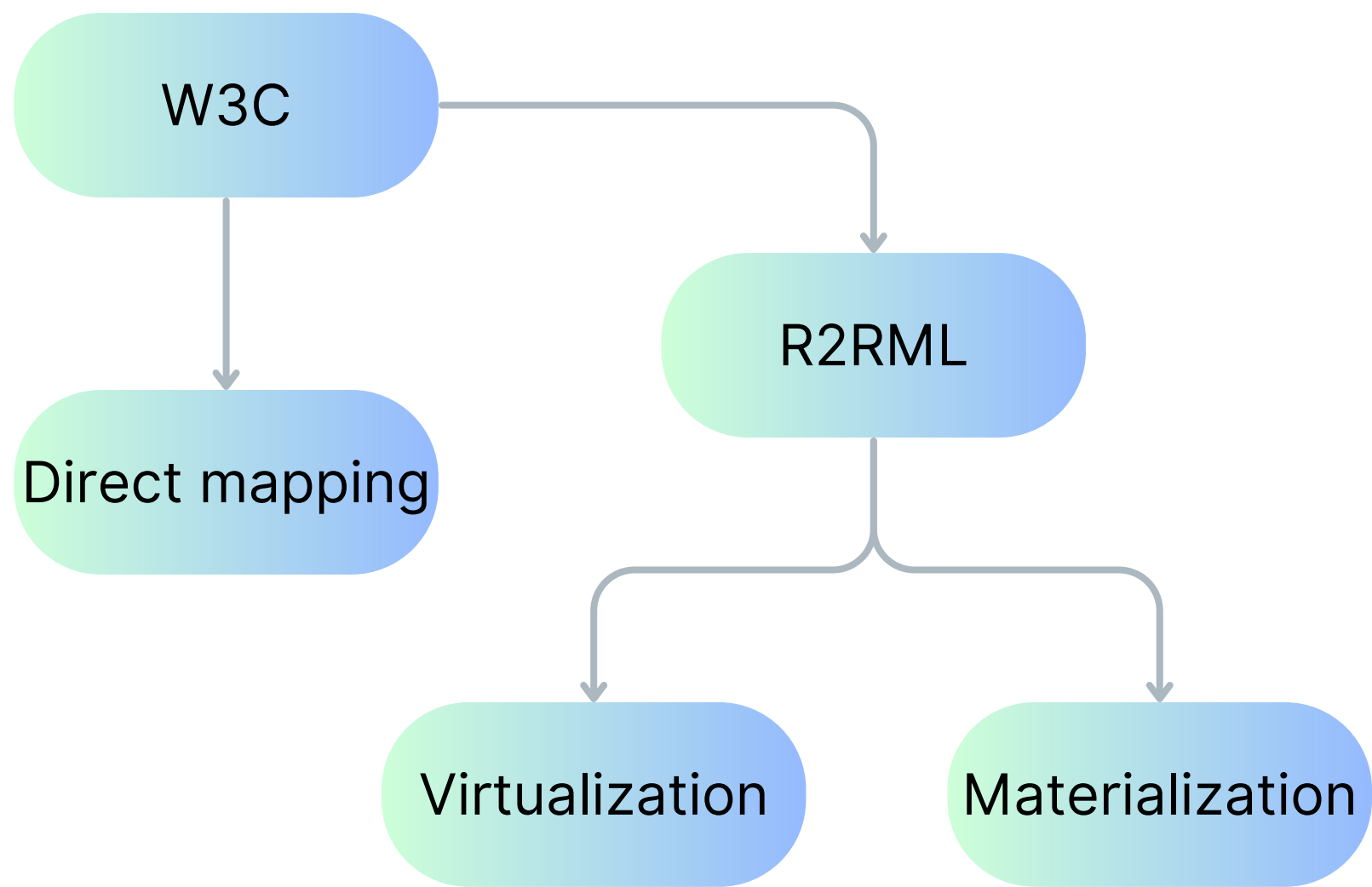
Database Element	Example	RDF Equivalent	RDF Example	
Table (Entity)	Patient, Doctor	Class	:Patient, :Doctor	# Doctor entities :Doctor_D101 a :Doctor ; :hasName "Dr. Smith" ; :hasDepartment "Oncology" . :Doctor_D102 a :Doctor ; :hasName "Dr. Brown" ; :hasDepartment "Pediatrics" .
Row (Record)	PatientID=1	Subject (Entity)	:Patient_1	
Column (Attribute)	Name, Age	Property	:hasName, :hasAge	# Patient entities :Patient_1 a :Patient ; :hasName "Alice" ; :hasAge "30" ; :treatedBy :Doctor_D101 . :Patient_2 a :Patient ; :hasName "Bob" ; :hasAge "45" ; :treatedBy :Doctor_D102 .
Foreign Key	DoctorID	Relationship (Object Property)	:treatedBy	
Cell Value	"Alice"	Object (Value)	"Alice"	
Foreign Key Value (Reference)	D101	Reference to another subject	:Doctor_D101	

TABLE TO RDF



RDF TRANSFORMATION FRAMEWORK

Method 2



	PostgreSQL		MySQL	
	passed	failed	passed	failed
Morph-RDB	27	35	31	31
Ontop	59	3	45	17
DB2Triples	13	49	24	38
R2RML-F	13	49	24	38

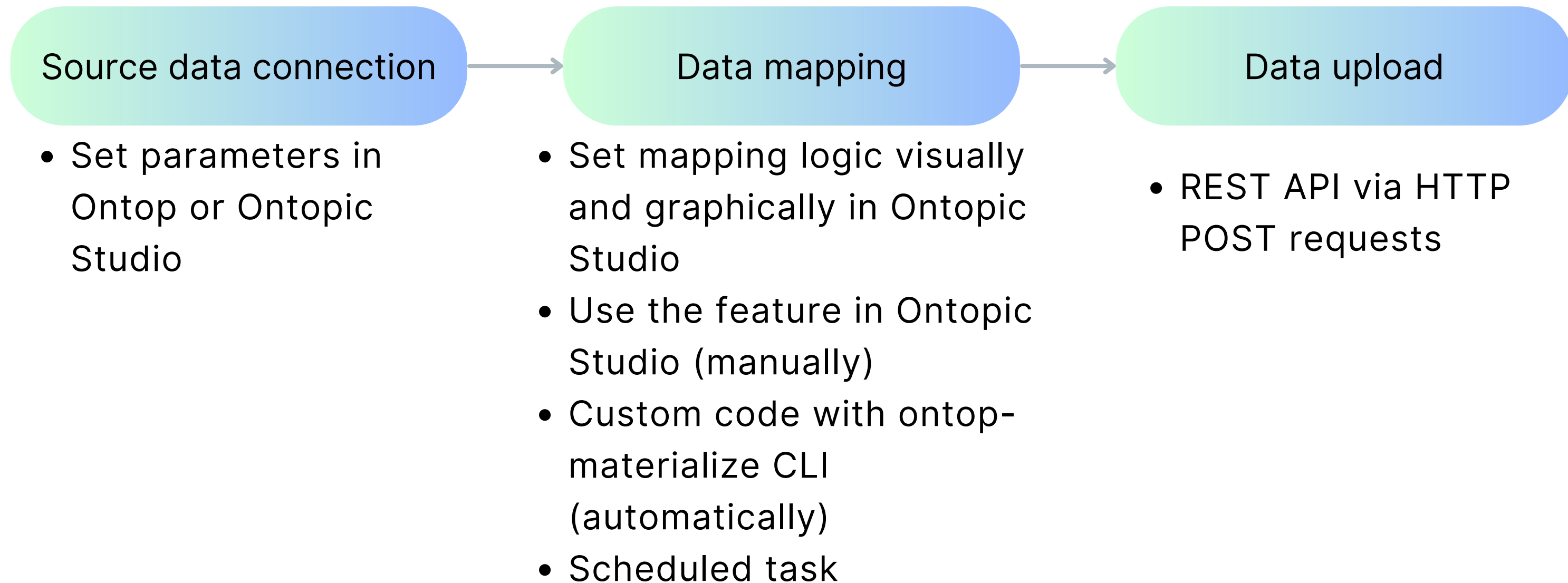
Comparison of R2RML engine

[1]<https://chrdebru.github.io/courses/ois/ois-lecture-07.pdf>
[2]Direct Mapping and Turtle Ontology for Management of Indonesian Movies Knowledge. MATEC Web of Conferences, 61372. <https://doi.org/10.1051/matecconf/202237204011>
[3]Knowledge graph construction with R2RML and RML: An ETL system-based overview. CEUR Workshop Proceedings, 2873. <https://ceur-ws.org/Vol-2873/paper11.pdf>

RDF TRANSFORMATION FRAMEWORK



Method 2



[1]<https://ontopic.ai/en/ontopic-studio/>

[2]<https://ontop-vkg.org/tutorial/basic/setup.html#database-setup>

[3]<https://ontop-vkg.org/tutorial/materialization/materialization.html#choosing-the-right-approach-for-your-use-case>

[4]<https://ontop-vkg.org/guide/cli.html#setup-ontop-cli>

[5]<https://franz.com/agraph/support/documentation/http-reference.html>